

AMARESH D R

AI Developer | Generative AI & ML Engineer | Data & Software Engineering

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PROFILE SUMMARY

AI Developer / AI-ML Engineer with 1+ year of professional, non-internship experience designing and building Machine Learning and Generative AI solutions across multimodal data (image, sensor, clinical, tabular, time-series). Skilled in the end-to-end ML solution lifecycle: data ingestion and pipeline automation, exploratory data analysis, feature engineering, model development (Scikit-learn, PyTorch), and Explainable AI (SHAP) for decision-support systems. Hands-on with deep learning architectures (CNN, Vision Transformers, LSTM, Transformers) and Generative AI / LLM techniques (GANs, VAEs, Diffusion Models, Large Language Models, Prompt Engineering, NLP), including diffusion-based data augmentation for imbalanced datasets. Strong foundation in Python, SQL, JavaScript, data structures, algorithms, and object-oriented design, with full-stack development experience building REST APIs (HTML, CSS, JavaScript, Django). Good understanding of cloud computing fundamentals and a fast learner with new AI/cloud tooling; works effectively in agile, cross-functional teams, translating ambiguous problem statements into production-grade, reproducible solutions.

CORE COMPETENCIES

- Machine Learning & Deep Learning Model Development
- Generative AI, LLMs & Prompt Engineering (GANs, VAEs, Diffusion Models)
- Data Pipeline Development, Automation & Multi-Source Ingestion at Scale
- Exploratory Data Analysis (EDA), Feature Engineering & Anomaly Detection
- Explainable AI (SHAP) & Model Validation for Decision Support
- Computer Vision (CNN, Vision Transformers) & NLP / Transformer Architectures
- Data Visualization & Dashboarding (Power BI, Tableau, Plotly)
- Full-Stack Web Development & REST API Design (Python, Django, JavaScript)
- Agile Software Development & Cross-Functional Collaboration
- Data Structures, Algorithms & Object-Oriented Design
- Technical Documentation & Reproducible Engineering

TECHNICAL SKILLS

- Programming Languages: Python, SQL, JavaScript, HTML, CSS
- Machine Learning: Scikit-learn, SVM, Model Validation, Hyperparameter Tuning
- Deep Learning: PyTorch, CNN, Vision Transformers (ViT), LSTM, Transformers
- Generative AI & LLM Concepts: Large Language Models (LLMs), Prompt Engineering, NLP, GANs, Variational Autoencoders (VAE), Diffusion Models
- Explainable AI & Computer Vision: SHAP, OpenCV
- Data Engineering: Automated Pipeline Development, Multi-Source Ingestion, Preprocessing, Feature Engineering
- Data Analysis & Visualization: Pandas, NumPy, SciPy, Matplotlib, Power BI, Tableau, Plotly
- Web Development: Python (Django), JavaScript, REST APIs, HTML5, CSS3
- Engineering Practices: Agile, Git, GitHub, Modular Code Design, Unit Testing/Validation, Technical Documentation
- Domain Exposure: CAN/LIN Vehicle Communication Protocols, Vehicle & Sensor Data Logging

PROFESSIONAL EXPERIENCE

Research Analyst / Technical Programmer — Researcher Lyceum Private Limited

Jul 2025 – Present

- Designed and coded scalable, automated data pipelines from broadly-defined problem statements to ingest, clean, and organize multimodal data (image, sensor, clinical, tabular) from multiple sources, reducing manual preprocessing effort for downstream ML systems.
- Built and validated machine learning models to run predictions across multi-source, high-volume datasets, performing exploratory data analysis to identify trends, anomalies, and performance characteristics that directly supported root-cause analysis workflows.
- Developed and shipped multimodal deep learning models for fetal health prediction by integrating imaging data with structured clinical records; applied Explainable AI (SHAP) to produce transparent, decision-support-ready outputs.
- Engineered a Generative AI-based defect detection framework for automated quality inspection using diffusion-based data augmentation and Vision Transformers (ViT), improving detection accuracy on imbalanced datasets.
- Implemented transformer-based architectures for materials science research, applying physics-aware modeling to structured/sensor data for stability prediction.
- Optimized model performance through hyperparameter tuning, model comparison, and architecture refinement using Python, SQL, PyTorch, Scikit-learn, OpenCV, Pandas, and NumPy within an agile, iterative delivery cycle.
- Collaborated with cross-disciplinary research and engineering teams across healthcare, manufacturing, and scientific domains, maintaining reproducible experimentation, clean code, and clear technical documentation.

Full Stack Developer Intern — Networkz Systems

May 2023 – Jul 2023

- Designed, coded, and shipped responsive full-stack web applications end-to-end (HTML, CSS, JavaScript front end; Python/Django back end with REST APIs) within a 2-person agile team, delivering features iteratively under tight timelines.
- Built foundational experience developing internal dashboards and automation tooling, working directly with teammates to manage ambiguity and fast-changing requirements.

PROJECTS

Generative AI–Based Defect Detection for Automated Quality Inspection

Built an automated visual defect/anomaly detection system using Vision Transformers (ViT) and diffusion-based data augmentation to address class imbalance and improve detection accuracy for industrial quality inspection — directly applicable to large-scale, component-level test analysis (batteries, converters, relays, and other electrical systems).

Multimodal Fetal Health Analysis using Explainable AI

Built a multimodal prediction system integrating imaging data with clinical/tabular records using deep learning models; applied SHAP to produce interpretable, decision-support-ready predictions and enhance model transparency.

Physics-Aware Transformer Framework for Spinel Stability Prediction

Developed a transformer-based framework applying physics-aware modeling and structured-data tokenization to predict material stability — demonstrating the ability to model physical/electrical system behavior from structured, sensor-style data at scale.

CERTIFICATIONS

- Machine Learning Intern — Cognifyz Technologies
- Full Stack Web Development — Networkz Systems

EDUCATION

Bachelor of Engineering (B.E.), Computer Science and Engineering — Anna University

CGPA: 8.00/10 | Year of Graduation: 2025